

CHRONIC DISEASE PROGRESSION PREDICTION SYSTEM

Abstract

Chronic diseases such as Diabetes, Alzheimer's Disease, and Parkinson's Disease are long-term medical conditions that require continuous monitoring and treatment. Predicting disease progression at an early stage can help healthcare professionals provide better treatment and improve patient outcomes. This project develops a machine learning-based Chronic Disease Progression Prediction System using patient health data. A Random Forest Classifier was used to predict disease stages based on demographic, lifestyle, and medical information. The model achieved an accuracy of 99.83% and was integrated into a Streamlit web application for real-time prediction.

1. Introduction

Chronic diseases are among the leading causes of health complications worldwide. Monitoring disease progression is important for effective treatment and management. Traditional methods rely heavily on clinical observations and periodic assessments. Machine Learning provides an efficient way to analyze patient data and identify patterns that indicate disease progression.

This project focuses on predicting disease stages using healthcare-related features such as age, disease type, medical history, biomarker score, medication dose, and heart rate.

2. Objectives

- To analyze chronic disease patient data.
- To perform data preprocessing and feature engineering.
- To develop a machine learning model for disease stage prediction.
- To evaluate model performance using classification metrics.
- To create a web-based prediction system using Streamlit.

3. Dataset Description

The dataset contains 3000 patient records collected for chronic disease analysis.

Features Used

- PatientID
- Date
- Age
- Gender
- Disease

- MedicalHistory
- Lifestyle
- BiomarkerScore
- MedicationDose
- HeartRate

Target Variable

DiseaseStage

The DiseaseStage feature was created using BiomarkerScore values and categorized into:

- Early Stage
- Mid Stage
- Late Stage

4. Data Preprocessing

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning.

The following steps were performed:

- Missing value detection
- Missing value treatment
- Duplicate record checking
- Label Encoding of categorical variables
- Feature Engineering
- Train-Test Split

Missing values in the MedicalHistory column were handled appropriately to ensure data consistency. Categorical features such as Gender, Disease, Lifestyle, and MedicalHistory were converted into numerical format using Label Encoding.

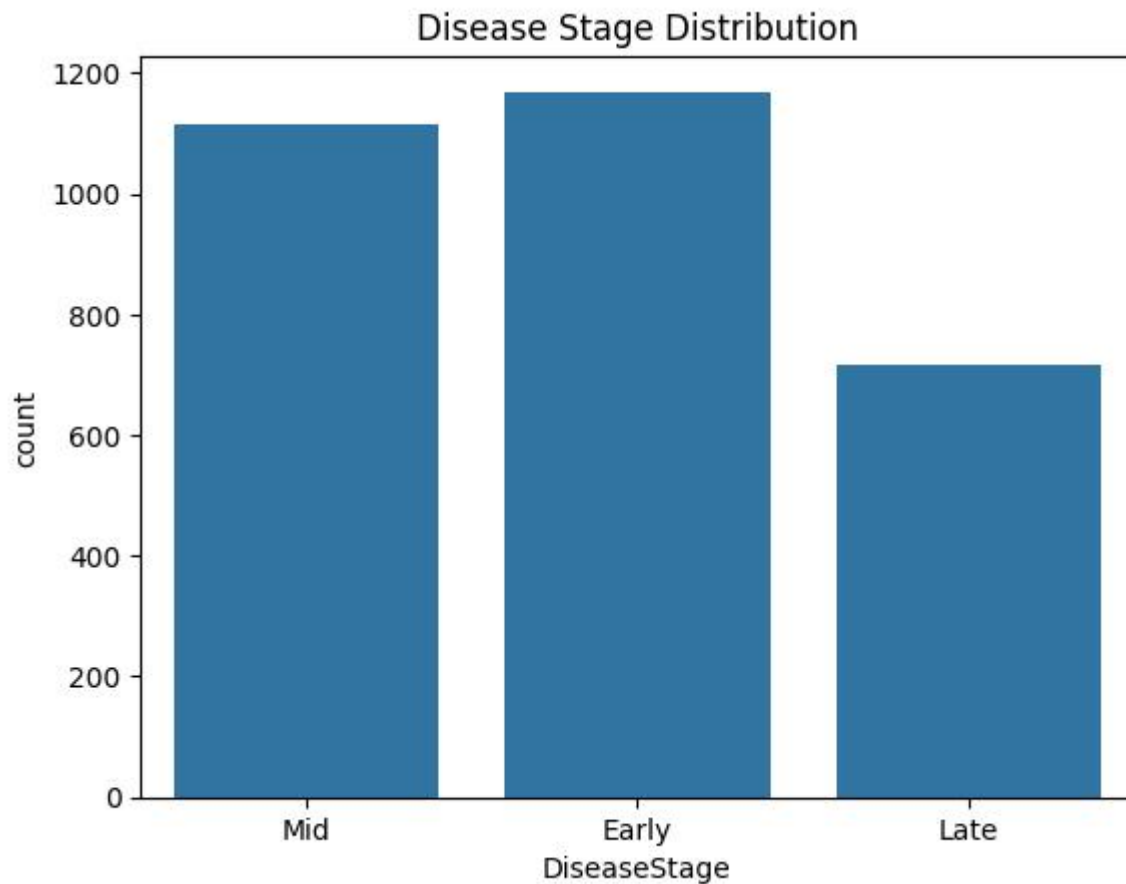
5. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the dataset and identify patterns among different features.

The analysis focused on:

- Disease Stage Distribution

- Age Distribution
- Biomarker Score Analysis
- Feature Correlation



Disease Stage Distribution Graph

The graph shows the distribution of Early, Mid, and Late disease stages. The data distribution is reasonably balanced, which helps improve model performance and reduces prediction bias.

6. Machine Learning Model

The Random Forest Classifier was selected for disease stage prediction.

Reasons for Choosing Random Forest

- High prediction accuracy
- Handles large datasets effectively
- Reduces overfitting
- Works well with healthcare data

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

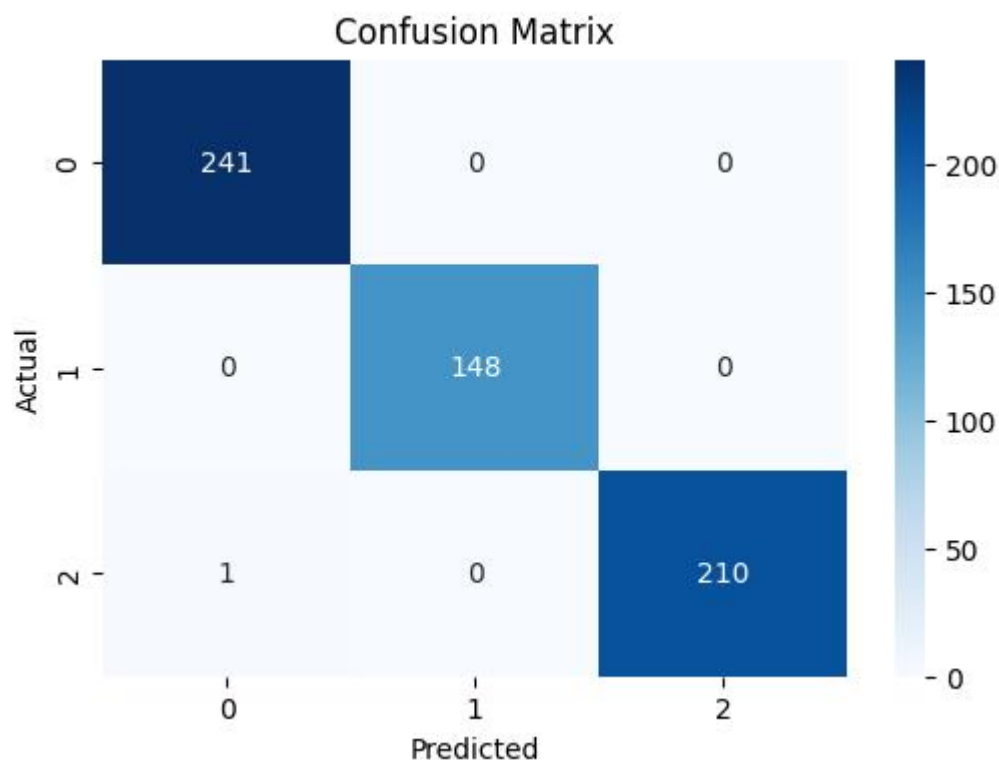
The model was trained using the processed dataset and evaluated using standard classification metrics.

7. Results and Evaluation

The trained Random Forest model demonstrated excellent predictive performance.

Performance Metrics

- Accuracy: 99.83%
- High Precision
- High Recall
- High F1-Score



Confusion Matrix

The confusion matrix indicates that most disease stages were classified correctly by the model. The low number of misclassifications demonstrates the effectiveness of the Random Forest algorithm in predicting disease progression stages.

Feature Importance

The most influential features were:

1. BiomarkerScore – 70.0%
2. Disease – 23.7%
3. Age – 1.8%
4. MedicationDose – 1.7%
5. HeartRate – 1.4%

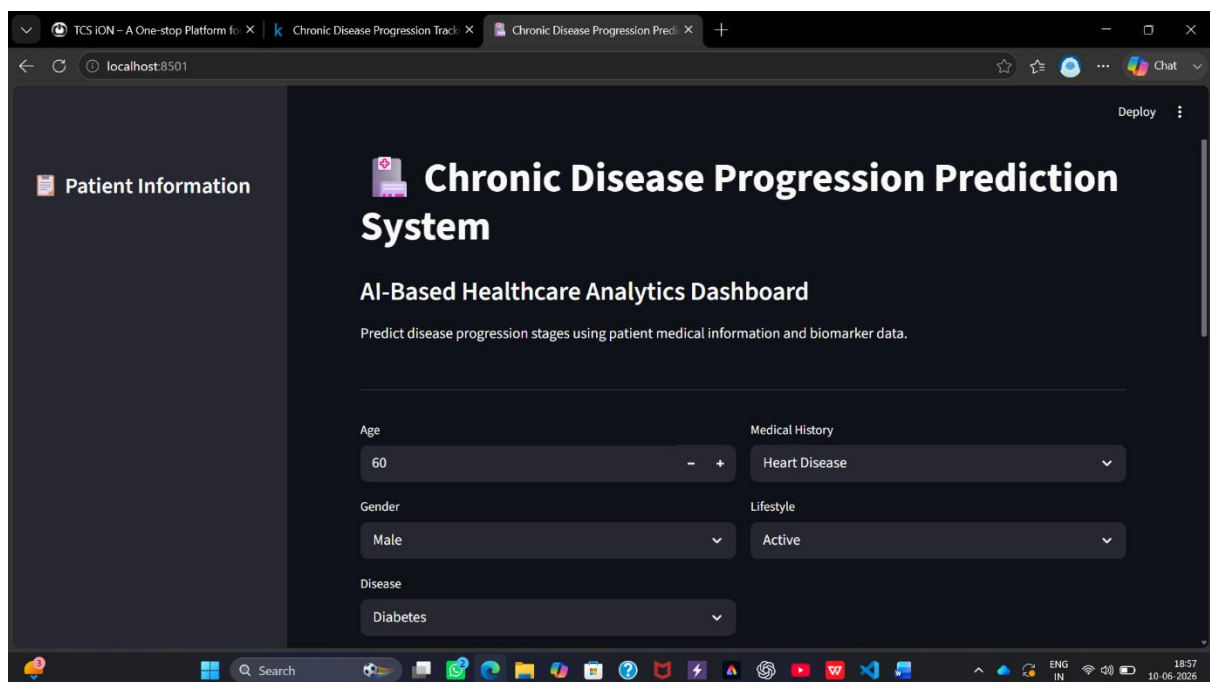
BiomarkerScore was identified as the most important feature affecting disease stage prediction.

8. Web Application Development

A web application was developed using Streamlit to provide real-time disease stage prediction.

Application Features

- User-friendly interface
- Patient data input
- Real-time prediction
- Fast and accurate results



Streamlit Web Application

The web application allows users to enter patient details and instantly obtain disease stage predictions. This makes the system accessible and practical for healthcare-related use cases.

9. Conclusion

This project successfully developed a Chronic Disease Progression Prediction System using Machine Learning techniques. The Random Forest Classifier achieved an accuracy of 99.83%, demonstrating excellent performance in disease stage prediction. The developed Streamlit application provides a simple and effective interface for real-time predictions.

10. Future Scope

- Integration with hospital databases
- Cloud deployment
- Real-time patient monitoring
- Advanced deep learning models
- Healthcare decision support systems

11. References

1. Scikit-Learn Documentation
2. Pandas Documentation
3. Streamlit Documentation
4. Machine Learning Research Papers
5. Healthcare Analytics Resources